RADINTEL AI / RADIOLOGY ONLY

Problem-Solving Use Cases

Kosovo radiology challenges, clinical workflows, proposed solutions, metrics and validation

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1. Purpose and scope

Identify radiology-specific operational problems in Kosovo and translate them into actionable, testable software use cases. Scope includes diagnostic imaging administration, DICOM/PACS integration, radiologist reporting, historical study access, and workflow analytics. Cardiology and autonomous diagnosis are explicitly excluded.

2. Evidence and problem framing

Kosovo is pursuing integrated digital health infrastructure through the KOMPAS program and the national Integrated Health Information System (IHIS). The World Bank has documented health-system information-sharing and capacity-building needs. This supports interoperability-first design; it does not prove that every radiology clinic has the same problems.

Issue	Status of evidence	Research needed
Fragmented information exchange	National digital-health modernization documented	Clinic-level RIS/PACS landscape
Imaging appointment delays	Reported in local media; varies by facility	Facility-specific queue and scanner capacity
Reporting workload	Plausible workflow bottleneck	Time-and-motion study with radiologists
Previous studies hard to locate	Hypothesis, not universal fact	Interview clinicians and IT staff
Equipment downtime	Facility-dependent hypothesis	Maintenance logs and service levels

3. Prioritization method

Score each proposed use case 1-5 on patient/clinic impact, frequency, implementability, procurement feasibility, and clinical/regulatory exposure. Start with features that can be demonstrated using synthetic or de-identified data, and whose output remains physician-controlled.

4. Detailed problem-solving use cases

UC-01: Structured radiology reporting

Problem: Radiologist spends time formatting repetitive reports.

Workflow: Doctor opens assigned study, selects template, dictates findings in Albanian/English, AI drafts structure, doctor checks against images and signs.

Measure: Report drafting time, edits per draft, signed-report turnaround.

Safety boundary: No invented findings; AI content always draft; physician is accountable.

UC-02: Retrieve prior imaging reports

Problem: Historical reports are scattered or difficult to search.

Workflow: Search by patient identifier and examination type; filter by date; display report timeline and related studies.

Measure: Median time to find prior report; retrieval success.

Safety boundary: Patient matching must be verified; role-based authorization.

UC-03: Compare textual findings across time

Problem: Radiologists manually review long historical report sets.

Workflow: Select two approved reports; system highlights additions, omissions and numeric changes with linked citations.

Measure: Comparison preparation time; clinician correction rate.

Safety boundary: Comparison is textual, not automated interpretation of pixels.

UC-04: Appointment and waiting-list visibility

Problem: Schedulers lack consistent operational overview.

Workflow: Book requested examination, record priority assigned by authorized staff, offer cancellations, audit changes.

Measure: No-show rate; time-to-appointment; scanner utilization.

Safety boundary: System never independently sets clinical urgency.

UC-05: Examination worklist

Problem: Work assignments and status updates are manual.

Workflow: Technician registers completed acquisition; study enters radiologist worklist; doctor claims, drafts, reviews, signs.

Measure: Unassigned studies; reporting turnaround.

Safety boundary: Avoid double assignment; emergency escalation defined by clinic.

UC-06: DICOM study viewer

Problem: Studies reside in separate PACS or local workstations.

Workflow: Connect authorized DICOMweb archive; open study in OHIF; view series, metadata and annotations.

Measure: Study open latency; failed retrievals.

Safety boundary: Diagnostic use requires validated display environment and clinical acceptance.

UC-07: Secure referring-doctor delivery

Problem: Reports are sent through ad-hoc channels.

Workflow: Generate approved report and time-limited access link; log each view/download; support secure institutional delivery.

Measure: Delivery latency; access incidents.

Safety boundary: No uncontrolled public links or exposure of patient identifiers.

UC-08: Quality and amendment workflow

Problem: Corrections can obscure report history.

Workflow: Signed report is immutable; issue an addendum with reason, signer and timestamps; notify relevant users.

Measure: Amendment traceability; audit completeness.

Safety boundary: No silent edits after signature.

UC-09: Operational dashboard

Problem: Managers cannot measure bottlenecks consistently.

Workflow: Aggregate modality volumes, backlog, report turnaround, cancellations and scanner downtime.

Measure: Weekly backlog; median turnaround; utilization.

Safety boundary: Use minimum necessary personal data in analytics.

UC-10: AI clinical knowledge retrieval

Problem: Staff need quick access to approved templates and policies.

Workflow: RAG answers questions only from curated, versioned institutional documents with citations.

Measure: Answer grounding rate; policy lookup time.

Safety boundary: Never substitute for clinical guidelines or clinician judgment.

UC-11: Cross-facility consultation

Problem: Specialists need second opinions across locations.

Workflow: Request consultation, share study/report to an authorized radiologist, record response and sign-off.

Measure: Consult turnaround; sharing errors.

Safety boundary: Consent/legal basis and identity checks before sharing.

UC-12: Equipment availability management

Problem: Downtime can cause schedule disruption.

Workflow: Record scanner downtime, service events and capacity; suggest nonclinical rescheduling options.

Measure: Unplanned downtime; rescheduled appointments.

Safety boundary: No direct control of medical devices.

5. End-to-end illustrative clinic journey

A referring clinician orders a chest CT. A scheduler registers the order and obtains the required information. A technician completes the examination on the scanner. The DICOM study is indexed in PACS and linked to the worklist. The radiologist reviews images in a clinically accepted viewer, uses a structured template and optional speech transcription, checks prior reports, finalizes and signs. The referring clinician receives a controlled-access result. Clinic management sees anonymized turnaround metrics.

6. Pilot hypotheses and validation

Hypothesis	Pilot test	Decision criterion
Drafting reduces admin time	Measure 30-50 reporting sessions with/without tool	Meaningful reduction with no material safety regression
Prior reports are slow to locate	Observe 20 authorized retrieval tasks	Improved median retrieval time
Worklist improves coordination	Track 4-week baseline and pilot	Fewer unassigned studies and missing statuses
Clinics will pay	Interview 10-15 clinics and run paid pilots	At least 2 paying pilot commitments

7. Explicit non-goals

- No autonomous diagnosis, clinical triage, or automated findings from raw pixels in MVP.
- No replacement of PACS, national health information infrastructure, or licensed clinical judgment.
- No clinical use with real patient data before security, legal, and clinical readiness gates.

8. Dependencies and open questions

- Confirm local clinic workflow, imaging modalities, RIS/PACS vendors, DICOMweb support, and network topology.
- Confirm Kosovo data-protection obligations, data residency requirements, contractual roles, and relevant medical-device classification with counsel.
- Recruit clinical advisor and 1-2 pilot clinics; define safe test datasets and approval responsibilities.

References and evidence boundaries

World Bank, Kosovo healthcare strengthening (2024) — https://www.worldbank.org/en/news/press-release/2024/05/07/world-bank-supports-enhancement-of-health-services-in-kosovo-improving-the-lives-of-its-people

Kosovo Ministry of Health, IHIS oversight procurement (2026) — https://msh.rks-gov.net/Documents/Index/1?year=2026

World Bank, Kosovo Digital Health Blueprint case study — https://thedocs.worldbank.org/en/doc/b929829fc3d36bebdf9d93555c64f222-0140022024/related/Digital-Health-Blueprint-Toolkit-Kosovo-Case-Study.pdf

HL7 FHIR specification — https://hl7.org/fhir/

OHIF documentation — https://docs.ohif.org/

OHIF DICOMweb and Orthanc integration — https://docs.ohif.org/configuration/datasources/dicom-web/

Market demand, exact waiting times, clinic counts, current vendor penetration, willingness to pay, and legal classification require local primary research. All numerical commercial scenarios in this document are illustrative assumptions.